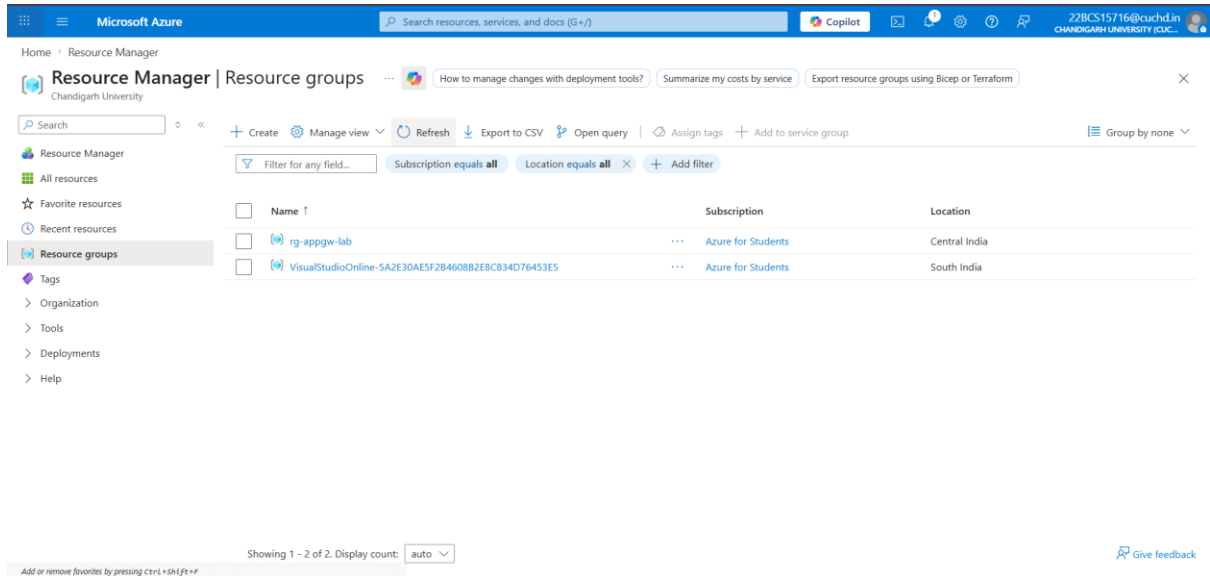


Azure Gateway CDN Exercise

STEP 1: Create Resource Group



Microsoft Azure

Search resources, services, and docs (G+I)

Copilot

22BCS15716@cuchd.in
CHANDIGARH UNIVERSITY (CUC...)

Home > Resource Manager

Resource Manager | Resource groups

How to manage changes with deployment tools? Summarize my costs by service Export resource groups using Bicep or Terraform

Search

Filter for any field...

Subscription equals all Location equals all Add filter

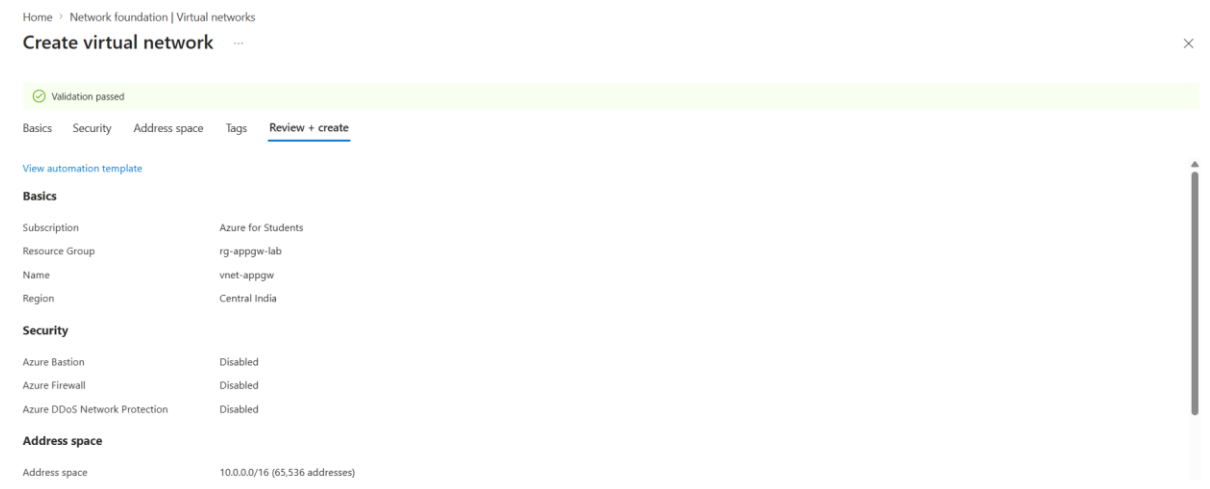
Name ↑	Subscription	Location
rg-appgw-lab	Azure for Students	Central India
VisualStudioOnline-5A2E30AE5F2B4608B2EBC834D76453E5	Azure for Students	South India

Showing 1 - 2 of 2. Display count: auto

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Give feedback

STEP 2: Create Virtual Network



Home > Network foundation | Virtual networks

Create virtual network

Validation passed

Basics Security Address space Tags Review + create

View automation template

Basics

Subscription	Azure for Students
Resource Group	rg-appgw-lab
Name	vnet-appgw
Region	Central India

Security

Azure Bastion	Disabled
Azure Firewall	Disabled
Azure DDoS Network Protection	Disabled

Address space

Address space	10.0.0.0/16 (65,536 addresses)
---------------	--------------------------------

Microsoft Azure

Search resources, services, and docs (G+J)

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Home > vnet-appgw-1775794977834 | Overview

vnet-appgw

Virtual network

Retrieve detailed routing information for troubleshooting

Suggest connectivity model for this network

Review flow metrics for my Virtual Network

Search

Move Delete Refresh Give feedback

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Resource visualizer

Settings

Address space

Connected devices

Subnets

Bastion

DDoS protection

Firewall

Microsoft Defender for Cloud

Network manager

DNS

Essentials

Resource group (move) : rg-appgw-lab

Location (move) : Central India

Subscription (move) : Azure for Students

Subscription ID : 5c9d415b-9260-4cf5-a6e3-b23b39d006d4

Address space : 10.0.0.0/16

Subnets : 1 subnet

DNS servers : Azure provided DNS service

BGP community string : Configure

Virtual network ID : d6f4c7da-ab6c-4756-a174-fb230850c598

Tags (edit) : Add tags

Topology Properties Capabilities Recommendations Tutorials

DDoS protection

Configure additional protection from distributed denial of service attacks.

Not configured

Azure Firewall

Protect your network with a stateful L3-L7 firewall.

Not configured

Peering

Seamlessly connect two or more virtual networks.

Not configured

Microsoft Defender for Cloud

Strengthen the security posture of your environment.

Private endpoints

Add or remove favorites by pressing Ctrl+Shift+F

Microsoft Azure

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Home > vnet-appgw-1775794977834 | Overview > vnet-appgw

vnet-appgw

Virtual network

Retrieve detailed routing information for troubleshooting

Suggest connectivity model for this network

Review flow metrics for my Virtual Network

Search

+ Subnet Refresh Manage users Delete

Create subnets to segment the virtual network address space into smaller address ranges.

Search subnets

Name ↑ IPv4

No items found

No subnets

Add a subnet

Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. Learn more

Subnet purpose : Default

Name : appgw-subnet

IPv4

Include an IPv4 address space : ☒

IPv4 address range : 10.0.0.0/16

Starting address : 10.0.0.0

Size : /24 (256 addresses)

Subnet address range : 10.0.0.0 - 10.0.0.255

IPv6

Include an IPv6 address space : ☐ This virtual network has no IPv6 address ranges.

Private subnet

Private subnets enhance security by not providing default outbound access. To enable outbound connectivity for virtual machines to access the internet, it is necessary to explicitly grant outbound access. A NAT gateway is the recommended way to provide outbound connectivity for virtual machines in the subnet. Learn more

Add Cancel

Give feedback

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Microsoft Azure

Search resources, services, and docs (G+J)

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Home > vnet-appgw-1775794977834 | Overview > vnet-appgw

vnet-appgw

Virtual network

Retrieve detailed routing information for troubleshooting

Suggest connectivity model for this network

Review flow metrics for my Virtual Network

Search

+ Subnet Refresh Manage users Delete

Create subnets to segment the virtual network address space into smaller address ranges.

Search subnets

Name ↑ IPv4

appgw-subnet 10.0.0.0/24

Showing 1 subnet

Add a subnet

Select an address space and configure your subnet. You can customize a default subnet or select from subnet templates if you plan to add select services later. Learn more

Subnet purpose : Default

Name : backend-subnet

IPv4

Include an IPv4 address space : ☒

IPv4 address range : 10.0.0.0/16

Starting address : 10.0.1.0

Size : /24 (256 addresses)

Subnet address range : 10.0.1.0 - 10.0.1.255

IPv6

Include an IPv6 address space : ☐ This virtual network has no IPv6 address ranges.

Private subnet

Private subnets enhance security by not providing default outbound access. To enable outbound connectivity for virtual machines to access the internet, it is necessary to explicitly grant outbound access. A NAT gateway is the recommended way to provide outbound connectivity for virtual machines in the subnet. Learn more

Add Cancel

Give feedback

Add or remove favorites by pressing Ctrl+Shift+F

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Settings
 - Address space
 - Connected devices
 - Subnets**
 - Bastion
 - DDoS protection
 - Firewall
 - Microsoft Defender for Cloud
 - Network manager
 - DNS

Add or remove favorites by pressing Ctrl+Shift+F

[+ Subnet](#) [Refresh](#) | [Manage users](#) [Delete](#) [Export to CSV](#)

Create subnets to segment the virtual network address space into smaller ranges for use by your applications. When you deploy resources into a subnet, Azure assigns the resource an IP address from the subnet.

☐	Name ↑	IPv4	IPv6	Available IPs	Delegated to	Security group	Route table		
☐	appgw-subnet	10.0.0.0/24	-	251	-	-	-	🗑️	🔗
☐	backend-subnet	10.0.1.0/24	-	251	-	-	-	🗑️	🔗

Showing 2 subnets

[Give feedback](#)

Successfully added subnet

Successfully added subnet 'backend-subnet' to virtual network 'vnet-appgw'.

STEP 3: Create Backend VMs (2 VMs)

Create a virtual machine ...

[Help me create a VM optimized for high availability](#)[Help me choose the right VM size for my workload](#)[Help me create a low cost VM](#)

⚠️ Changing Basic options may reset selections you have made. Review all options prior to creating the virtual machine.

[Basics](#) [Disks](#) [Networking](#) [Management](#) [Monitoring](#) [Advanced](#) [Tags](#) [Review + create](#)

Create a virtual machine that runs Linux or Windows. Select an image from Azure marketplace or use your own customized image. Complete the Basics tab then Review + create to provision a virtual machine with default parameters or review each tab for full customization. [Learn more](#)

📘 This subscription may not be eligible to deploy VMs of certain sizes in certain regions.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *
Resource group *
[Create new](#)

Instance details

Virtual machine name *

[< Previous](#) [Next: Disks >](#) [Review + create](#)[Give feedback](#)[Help me copy this VM in any region](#)[Manage this VM with Azure CLI](#)

- Overview**
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Monitor
- Resource visualizer
- Connect
 - Connect
 - Bastion
- Networking
 - Network settings
 - Load balancing
- Application security groups
- Network manager
- Settings

Add or remove favorites by pressing Ctrl+Shift+F

[Help me copy this VM in any region](#)[Connect](#) [Start](#) [Restart](#) [Stop](#) [Hibernate](#) [Capture](#) [Delete](#) [Refresh](#) [Scale](#) [Open in mobile](#) [Feedback](#) [CLI / PS](#)

Essentials

Resource group [\(move\)](#) : rg-appgw-lab
Status : Running
Location : Central India
Subscription [\(move\)](#) : Azure for Students
Subscription ID : 5c9d415b-9260-4cf5-a6e3-b23b39d006d4

Operating system : Linux (ubuntu 24.04)
Size : Standard_DC1s_v3
Primary NIC public IP : [52.172.138.11](#)
[1 associated public IPs](#)
Virtual network/subnet : [vnet-appgw/backend-subnet](#)
DNS name : [Not configured](#)
Health state : -
Time created : 10/4/2026, 5:03 am UTC

[JSON View](#)[Tags \(edit\)](#) [Add tags](#)[Properties](#) [Monitoring](#) [Capabilities \(7\)](#) [Recommendations](#) [Tutorials](#)

Virtual machine

Computer name : vm1
Operating system : Linux (ubuntu 24.04)
VM generation : V2
Agent status : Ready

Networking

Public IP address
[1 associated public IPs](#)
Public IP address (IPv6) : -
Private IP address : 10.0.1.4

Microsoft Azure

Home > CreateVm-canonical.ubuntu-24_04-lts-server-20260410103440 | Overview

vm2 Virtual machine

Help me copy this VM in any region

Search

Help me copy this VM in any region

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Monitor

Resource visualizer

Connect

Connect

Bastion

Networking

Network settings

Load balancing

Application security groups

Network manager

Settings

Add or remove favorites by pressing Ctrl+Shift+F

Essentials

Resource group (move) : rg-appgw-lab

Status : Running

Location : Central India

Subscription (move) : Azure for Students

Subscription ID : Scl9d415b-9260-4cf5-a6e3-b23b39d006d4

Tags (edit) : Add tags

Operating system : Linux (ubuntu 24.04)

Size : Standard DC1s v3 (1 vcpu, 8 GiB memory)

Primary NIC public IP : 4.188.248.243

1 associated public IPs

Virtual network/subnet : vnet-appgw/backend-subnet

DNS name : Not configured

Health state : -

Time created : 10/4/2026, 5:06 am UTC

Properties

Monitoring

Capabilities (7)

Recommendations

Tutorials

Virtual machine

Computer name : vm2

Operating system : Linux (ubuntu 24.04)

VM generation : V2

VM architecture : x64

Networking

Public IP address : 4.188.248.243 (Network interface vm2407)

1 associated public IPs

Public IP address (IPv6) : -

Private IP address : 10.0.1.5

Microsoft Azure

Home > Compute infrastructure

Compute infrastructure | Virtual machines

Check which VMs cannot access the internet

Clone a selected virtual machine

List all alerts for virtual machines

Search

Virtual machines

Get started

Create

Reservations

Manage view

Refresh

Export to CSV

Open query

Assign tags

Start

Restart

Stop

Group by none

Filter for any field

Subscription equals all

Type equals all

Resource Group equals all

Location equals all

Add filter

Name	Subscription	Resource Group	Location	Status	Operating syst...	Size	Public IP addre...	Disks
vm1	Azure for Stud...	rg-appgw-lab	Central India	Running	Linux	Standard_DC1s...	52.172.138.11	1
vm2	Azure for Stud...	rg-appgw-lab	Central India	Running	Linux	Standard_DC1s...	4.188.248.243	1

Showing 1 - 2 of 2. Display count: auto

Give feedback

Add or remove favorites by pressing Ctrl+Shift+F

Install Web Server For Ubuntu:

```

Administrator: Windows Command Prompt
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc ssl-cert
The following NEW packages will be installed:
  nginx nginx-common
0 upgraded, 2 newly installed, 0 to remove and 34 not upgraded.
Need to get 565 kB of archives.
After this operation, 1596 kB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx-common all 1.24.0-2ubuntu7.6 [43.5 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu noble-updates/main amd64 nginx amd64 1.24.0-2ubuntu7.6 [521 kB]
Fetched 565 kB in 0s (11.5 MB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 69217 files and directories currently installed.)
Preparing to unpack .../nginx-common_1.24.0-2ubuntu7.6_all.deb ...
Unpacking nginx-common (1.24.0-2ubuntu7.6) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx_1.24.0-2ubuntu7.6_amd64.deb ...
Unpacking nginx (1.24.0-2ubuntu7.6) ...
Setting up nginx-common (1.24.0-2ubuntu7.6) ...
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service.
Setting up nginx (1.24.0-2ubuntu7.6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@vm1:~# echo "Hello from VM1" | sudo tee /var/www/html/index.html
Hello from VM1
root@vm1:~#
Broadcast message from root@vm1 (Fri 2026-04-10 06:31:36 UTC):

root@vm2:~#
Broadcast message from root@vm2 (Fri 2026-04-10 06:31:36 UTC):
  
```

STEP 4: Create Application Gateway

• Basics Tab

Home > Load balancing and content delivery | Application gateways

Create application gateway ...

⚠ Changes you make on this tab may affect any configuration you've done on other tabs. Review all options prior to creating the application gateway.

✓ Basics ✓ Frontends ✓ Backends ④ Configuration ⑤ Tags ⑥ Review + create

An application gateway is a web traffic load balancer that enables you to manage traffic to your web application. [Learn about creating application gateway](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *	Azure for Students
Resource group *	rg-appgw-lab

[Create new](#)

Instance details

Application gateway name *	appgw-demo
Region *	Central India
Tier	Standard V2
Enable autoscaling	☒ Yes ☐ No

[Previous](#)

[Next : Frontends >](#)

• Frontend IP

✓ Basics ✓ Frontends ✓ Backends ④ Configuration ⑤ Tags ⑥ Review + create

Traffic enters the application gateway via its frontend IP address(es). An application gateway can use a public IP address, private IP address, or one of each type.

Frontend IP address type ☒ Public ☐ Private ☐ Both

Public IPv4 address *	(New) appgw-pip
-----------------------	-----------------

[Add new](#)

• Backend Pool

✓ Basics ✓ Frontends ✓ **Backends** ④ Configuration ⑤ Tags ⑥ Review + create

A backend pool is a collection of resources to which your application gateway can send traffic. A backend pool can contain virtual machines, virtual machine scale sets, app services, IP addresses, or fully qualified domain names (FQDN). ↗

[Add a backend pool](#)

Backend pool	Targets	
backend-pool	∨ 2 targets	...
	vm1204	...
	vm2407	...

• Configuration Tab

Add a routing rule

×

Configure a routing rule to send traffic from a given frontend IP address to one or more backend targets. A routing rule must contain a listener and at least one backend target.

Rule name *

rule1

✓

Priority * ⓘ

100

✓

* Listener * Backend targets

A listener “listens” on a specified port and IP address for traffic that uses a specified protocol. If the listener criteria are met, the application gateway will apply this routing rule. ↗

Listener name * ⓘ

listener

✓

Frontend IP * ⓘ

Public IPv4

∨

Protocol ⓘ

☒ HTTP ☐ HTTPS ☐ TCP ☐ TLS

Port * ⓘ

80

✓

Listener type ⓘ

☒ Basic ☐ Multi site

Custom error pages

Show customized error pages for different response codes generated by Application Gateway. This section lets you configure Listener-specific error pages. [Learn more](#) ↗

Please verify that the url(s) being added here is reachable from your application gateway using the [connection troubleshoot](#) tool to prevent any deployment error.

Bad Gateway - 502

Enter Html file URL

Add

Cancel

* Listener * Backend targets

Choose a backend pool to which this routing rule will send traffic. You will also need to specify a set of Backend settings that define the behavior of the routing rule. ⓘ

Target type ☒ Backend pool ☐ Redirection

Backend target * ⓘ [Add new](#)

Backend settings * ⓘ [Add new](#)

Path-based routing

You can route traffic from this rule's listener to different backend targets based on the URL path of the request. You can also apply a different set of Backend settings based on the URL path. ⓘ

Path based rules

Path	Target name	Backend setting name	Backend pool
No additional targets to display			

Home > Load balancing and content delivery | Application gateways

Load balancing and content delivery ... Preview

Search

Overview

Load balancing

Application gateways

Load balancers

Content delivery

DNS load balancing

Related services

Create Manage view

Name ↑

appgw-demo

Showing 1 - 1 of 1. Display count: auto

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appgw-demo Application gateway

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Help

Essentials

Resource group (move) rg-appgw-lab

Location Central Ind Copy to clipboard

Subscription (move) Azure for Students

Subscription ID 5c9d415b-9260-4cf5-a6e3-b23b39d006d4

Tags (edit) Add tags

Recommendations Metrics

Virtual network/subnet vnet-appgw/appgw-subnet

Frontend public IP address 4.187.234.131 (appgw-pip)

Frontend private IP address -

Tier Standard V2

Availability zone 1, 2, 3

✓

Your resource is following Light best practices.

Azure Advisor provides personalized recommendations to reduce costs, increase security, ...

Add or remove favorites by pressing Ctrl+L+Shift+F+P

STEP 5: Test Application Gateway

Hello from VM1

Hello from VM2